

Drone Designer — Component Selection Engine

MTOW Iterator & Selection Logic Reference
Teknofest 2025 · Core/Services/ComponentSelectionEngine.vb

1 MTOW Fixed-Point Iterator

Battery mass and MTOW are mutually dependent — a heavier drone needs more power, which requires a larger battery, which increases MTOW. The solver resolves this circular dependency by iterating until the MTOW delta falls below 1 g.

1.1 Structural Mass (constant across all iterations)

```
m_struct = m_airframe + 35 (FC stack) + 10 (Rx) + 20 (GPS) + 30 (wiring) + m_payload
```

Motor count	3	4	6	8	12
Airframe mass (g)	350	250	420	680	1 200

1.2 Initial Battery Mass Seed

The seed fraction is chosen by endurance bracket, then scaled inversely with battery pack specific energy so that higher-density chemistries start with a smaller seed:

```
f_base = 0.25 if t < 15 min
f_base = 0.35 if 15 ≤ t ≤ 45 min
f_base = 0.45 if t > 45 min
f_seed = clamp( f_base × (130 / E_pack_Wh_kg), 0.10, 0.60 )
b_0 = m_struct × f_seed [grams]
```

Chemistry	Default E_pack (Wh/kg)	f_seed @ 24 min (base 0.35)
LiPo	130	0.35
Li-Ion	180	0.25
H■ Fuel Cell	350	0.13 (clamped → 0.13)
Hybrid FC+LiPo	240	0.19
Tethered	0	0.00 (no battery)

1.3 Iteration Loop

Each pass i recomputes MTOW, hover power, total energy, and the next battery mass:

```
MTOW_i = ( m_struct + b_i ) × 1.10 [+10 % safety]
P_hover = MTOW_i × 10 [W / 100 g] [empirical for 5"-7" quads]
E_total = P_hover × t_h × 1.20 [+20 % non-hover overhead]
E_usable = E_pack_Wh_kg × DoD [104 Wh/kg for LiPo @ 80 % DoD]
b_{i+1} = ( E_total / E_usable ) × 1000 [Wh → grams]
```

Convergence criterion: | MTOW_i – MTOW_{i-1} | ≤ 1 g

1.4 Divergence & Non-Convergence Guards

Divergence throw: if battery mass has increased for 3 consecutive passes AND $b_i > 1.5 \times b_0 \rightarrow$ mission is physically infeasible.

Non-convergence throw: if 10 iterations complete without reaching the 1 g tolerance $\rightarrow$ mission is at the edge of feasibility.

1.5 Worked Example — Default UI Values (quad, 0.4 h, 0 g payload, LiPo)

Inputs: N=4, t=0.4 h (24 min), m_payload=0 g, LiPo 130 Wh/kg

Pass	b_i (g)	MTOW (g)	P_hover (W)	E_total (Wh)	b_{i+1} (g)	Δ MTOW (g)
seed	121	—	—	—	—	—
1	121	512	51.2	24.6	236	512
2	236	639	63.9	30.7	295	127
3	295	704	70.4	33.8	325	65
4	325	737	73.7	35.4	341	33
5	341	754	75.4	36.2	348	17
6	348	762	76.2	36.6	352	8
7	352	766	76.6	36.8	354	4
8	354	768	76.8	36.9	355	2
9	355	769	76.9	37.0	356	≈1 ✓

* $b_0 = 121$ g. Divergence threshold = $1.5 \times 121 = 181$ g. b grows past 181 g at pass 2 (236 g) — guard fires if 3 consecutive increases have already occurred. Passes 1–3 are all increases $\rightarrow$ guard triggers at end of pass 3. This is the current bug: the guard is too aggressive for small drones where the seed is far below the true battery mass.

2 Thrust Requirement (Step 2)

$F_{total} = MTOW \times TWR$ $TWR = 2.0$ (all non-racing missions)

$F_{motor} = F_{total} / N$ [gram-force per motor]

Units are gram-force (gf) throughout. Motor datasheets rate static thrust in gf, avoiding unit conversion during filtering.

3 Propeller-First Selection Pipeline

3.1 Frame Geometry $\rightarrow$ Target Propeller Diameter (Step 3a)

$arm_length = diagonal / \sqrt{2}$

$max_prop_radius = arm_length / 2.0$ [FrameArmToPropRadiusRatio = 2.0]

$max_prop_diameter = 2 \times max_prop_radius$ [mm $\rightarrow$ inches: $\div 25.4$]

$target_diameter = max_prop_diameter - 0.5$ in [inset for catalog fit]

Motor count	Diagonal (mm)	Arm (mm)	Max prop diam (in)	Target (in)
3	220	155.6	12.2	11.7

4	250	176.8	13.9	13.4
6	380	268.7	21.2	20.7
8	500	353.6	27.8	27.3
12	700	494.9	38.9	38.4

Note: these target diameters (13–38 in) are computed correctly from the geometry but appear large relative to typical catalog props (5–10 in for small quads). The FrameArmToPropRadiusRatio=2.0 and diagonal table are calibrated for commercial/inspection class frames, not 250 mm racing frames.

3.2 Propeller Selection (Step 3b)

Hard filters:

```
prop.diameter ≤ max_prop_diameter [frame clearance – hard ceiling]
| prop.diameter – target_diameter | ≤ 1.5 in [tolerance band]
```

Ranking: distance from target ASC, then static thrust DESC. Take top 5.

Fallback: if zero candidates pass the tolerance band, drop the ±1.5 in filter and keep only the frame-clearance constraint. If still zero, throw.

3.3 Propeller Hover RPM (momentum theory)

$$\text{RPM}_{\text{hover}} = \text{RPM}_{\text{test}} \times \sqrt{(F_{\text{motor}} / F_{\text{static_at_test}})} \quad [T \propto \text{RPM}^2]$$

Fallback when no test data: $\text{RPM}_{\text{hover}} = \text{prop.MaxRPM} \times 0.70$

3.4 Motor Selection (Step 3c — propeller-first filters)

Hard filters (all must pass):

```
motor.PropDiameterMin ≤ prop.diameter ≤ motor.PropDiameterMax
motor.V_min ≤ V_nominal ≤ motor.V_max
motor.MaxThrust ≥ F_motor
motor.KV × V_nominal ≥ RPM_hover × 1.4 [RPM headroom factor]
```

Ranking: efficiency (gf/W) DESC, then mass ASC. Take top 5.

4 Power Budget (Step 4)

4.1 Motor Current

```
I_peak = motor.MaxCurrentA [datasheet – preferred]
= motor.MaxPowerW / V_nominal [fallback 1: P = VI]
= F_motor / 10 [fallback 2: ~1 A per 10 gf]

r = clamp( F_motor / motor.MaxThrust, 0.05, 1.0 )
I_hover = I_peak × r^1.5 [momentum theory: I ∝ T^1.5]
```

Fallback if motor.MaxThrust = 0: $I_{\text{hover}} = I_{\text{peak}} \times 0.35$ (legacy flat fraction).

4.2 System Current

```
I_sys_peak = I_peak × N + 3.0 A [3.0 A fixed avionics estimate]
```

```
I_sys_avg = I_hover × N + 3.0 A
```

4.3 Battery Capacity

```
C_min = ( I_sys_avg × t_h × 1000 ) / 0.80 [mAh, 80 % DoD]
```

```
C_req = C_min × 1.20 [+20 % margin]
```

4.4 C-Rating

```
C_rate_required = ( I_sys_peak / C_req_Ah ) × 1.15 [+15 % headroom]
```

5 Battery Selection (Step 5)

Hard filters (all must pass):

```
battery.cells = power.CellCount [exact match]
```

```
battery.capacity_mAh ≥ C_req
```

```
battery.C_rating ≥ C_rate_required
```

```
battery.mass_g ≤ MTOW_battery_estimate × 1.15 [+15 % mass tolerance]
```

Ranking: |capacity – C_req| ASC (closest fit), C_rating DESC, mass ASC. Take top 3.

6 ESC & PDB Sizing (Steps 6–7)

```
I_ESC_required = I_peak_motor × 1.25 [+25 % thermal margin]
```

```
I_PDB_required = I_sys_peak × 1.30 [+30 % wiring margin]
```

ESC filter: continuous rating ≥ I_ESC_required, max voltage ≥ V_nominal, not all-in-one. PDB filter: continuous rating ≥ I_PDB_required, max voltage ≥ V_nominal, ESC pad count ≥ N.

7 Diagnosis — Current Failure at Default UI Values

With defaults of 0.4 h endurance, 4-motor frame, 0 g payload, LiPo chemistry:

Parameter	Value
m_struct	345 g (250 airframe + 95 electronics)
b_0 (seed @ 24 min bracket)	121 g (345 × 0.35)
Divergence threshold	182 g (121 × 1.5)
b after pass 1	236 g — exceeds 182 g
Consecutive increases at pass 3	3 → guard fires
Guard verdict	THROW — 'MTOW diverging'

Root cause: The seed fraction ($0.35 \times m_{\text{struct}} = 121 \text{ g}$) severely underestimates the battery needed for a 0.4 h quad, which actually converges to ~355 g. The divergence guard interprets the expected growth as divergence and throws prematurely. The iterator is correct — the guard threshold ($1.5 \times \text{seed}$) is too tight when the seed is far from the true fixed point.

